

Inverse-Variance Proximal Policy Optimization (IV-PPO)

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Abstract—Standard Proximal Policy Optimization (PPO) treats all training samples equally, which limits sample efficiency and robustness in stochastic control tasks. This paper extends Inverse-Variance Reinforcement Learning (IV-RL) to PPO, introducing IV-PPO, which uses variance heads and target ensembles to weight updates by per-sample uncertainty. Tested on OpenAI Gym dense-reward control environments, IV-PPO improves sample efficiency specifically in heteroscedastic environments while maintaining baseline performance on homoscedastic tasks. On LunarLanderContinuous-v2, which exhibits state-dependent wind noise, IV-PPO achieves a 21.9% improvement over vanilla PPO and 6.2% over EnsemblePPO. Results on MuJoCo locomotion tasks (HalfCheetah-v4, Walker2d-v4) reveal that benefits depend on the nature of environmental stochasticity, with contact dynamics showing limited or negative transfer.

Index Terms—uncertainty estimation, sample efficiency, Proximal Policy Optimization (PPO), Reinforcement Learning (RL)

I. INTRODUCTION

Real-world deep reinforcement learning (DRL) environments often involve highly noisy targets and inherent stochasticity that can destabilize an agent’s learning [1]. Standard DRL algorithms like Deep Q-Network (DQN) and Soft Actor-Critic (SAC) do not account for this variability in target quality. Likewise, the popular PPO algorithm trains its value function (critic) by simple mean-squared error (MSE) regression to observed returns [2, 3], effectively treating all transitions as equally trustworthy. In highly stochastic environments, this can render PPO susceptible to instability or even overfitting to noise - for example, policy updates might chase random reward fluctuations or unreliable advantages, leading to erratic learning behaviour [4, 5]. Thus, a core problem is how to learn efficiently and stably in random environments by reducing the detrimental impact of noisy feedback on the learning process.

Inverse-Variance Reinforcement Learning (IV-RL) was recently proposed as a solution to this problem [1]. This is a Bayesian framework that explicitly quantifies the uncertainty in the temporal-difference (TD) target, so one can improve sample efficiency by down-weighting high-variance (uncertain) samples and trusting low-variance (reliable) samples more. IV-RL combines two complementary uncertainty estimation techniques in a loss function: (1) a variance head in the critic network to estimate the aleatoric uncertainty (outcome noise) of the TD target, and (2) an ensemble of critic networks

(target critics) to estimate epistemic uncertainty in the agent’s own value predictions. In the original paper, this approach was applied to off-policy methods (IV-DQN and IV-SAC) and demonstrated significantly faster and more stable learning by mitigating noisy TD supervision through inverse-variance sample reweighting.

In this paper, the IV-RL framework is generalized to PPO, an uncertainty-aware on-policy actor-critic method, to introduce IV-PPO. IV-PPO is empirically evaluated on two dense-reward environments with stochastic dynamics, comparing against standard and ensemble PPO as baselines. The evaluation centers on two criteria: (1) sample efficiency - how quickly the agent learns an effective policy in terms of interaction steps; and (2) learning stability - the consistency of training progress across time and random seeds (avoiding divergence or high variance in performance). This uncertainty-weighted learning strategy aims to make learning more sample-efficient and stable, as the agent focuses on the most informative experiences.

II. BACKGROUND & RELATED WORK

A. Stochasticity and Heteroscedasticity in TD Targets

In RL, value function updates are driven by TD targets computed from experience rollouts. A common example is the one-step estimate $T(s, a) = r + \gamma \bar{Q}(s', a')$, which inherits stochasticity from various sources: random rewards, probabilistic transitions, stochastic action selection, and approximation errors in the value function. Crucially, this noise is not uniform across samples - some state-action pairs consistently yield reliable returns, while others produce highly variable outcomes due to uncertainty in the environment or the model itself. This state- and sample-dependent variability is referred to as heteroscedasticity [1, 6-8].

Beyond static randomness in the environment, heteroscedasticity also emerges as a consequence of the agent’s learning process [6]. As training progresses, both the policy and value function evolve, causing the agent to visit different regions of the state space with different return distributions and function approximation quality. Consequently, the noise in TD targets becomes time-varying, depending on where the agent is and how well the critic generalizes in that region. This dynamic effect holds across both short-horizon TD targets and long-horizon return estimates (e.g., multi-step bootstrapping or

Monte Carlo returns), as these aggregate uncertainty across entire trajectories.

Despite this, most standard RL algorithms (e.g., DQN, PPO, SAC) treat all samples equally by minimizing unweighted value losses - implicitly assuming homoscedasticity. This means that highly noisy targets are given the same influence on learning as clean, reliable ones.

Recent work in areas like offline RL and constrained policy optimization has begun to explicitly account for heteroscedasticity [7]. By modifying learning objectives (e.g., in conservative Q-learning) to reflect non-uniform target reliability, these methods demonstrate that ignoring target variance can lead to suboptimal exploration, instability, and inefficient use of data.

B. Sources of Uncertainty in DRL

Uncertainty in DRL systems is typically divided into two types: aleatoric and epistemic uncertainty [1, 9-15]. Aleatoric uncertainty comes from the environment itself - randomness that cannot be reduced by collecting more data or improving the model. In contrast, epistemic uncertainty arises from limited knowledge: it captures the agent's uncertainty due to insufficient data or limited model capacity and can often be reduced with more exploration or better function approximators.

Prior work in DRL has sought to separate and quantify these uncertainties using methods like deep ensembles [1, 9], Bayesian inference [10, 11, 12, 15], Monte Carlo dropout [9, 10], deep kernel learning [9], and evidential networks [9, 13, 14]. This disentanglement allows for targeted improvements: epistemic uncertainty can guide exploration (e.g., via optimism or risk sensitivity), while aleatoric uncertainty can help quantify unavoidable risks in learned policies, improving robustness and generalization.

The IV-RL framework leverages both types of uncertainty to tackle heteroscedastic noise: In DRL, (1) aleatoric uncertainty arises as the inherent stochasticity in the TD target, reflecting variability from environment dynamics and reward distributions. For instance, even under a fixed policy, repeating the same state-action pair can yield different next states or rewards. This uncertainty is captured by a variance network (var-network) which outputs both the predicted mean Q-value and its variance, thus learning to quantify heteroscedastic noise in the return.

On the other hand, (2) epistemic uncertainty reflects how confident the agent is in its critic's value estimates due to limited data or function approximation errors. It is captured using a variance ensemble - an ensemble of var-networks trained with different initializations and data orderings - where the disagreement in Q-value predictions across the ensemble members reflects epistemic uncertainty. High epistemicity signals the need for more data or caution. Variance ensembles yield a Gaussian mixture variance combining environment stochasticity and model uncertainty, which decreases with more experience. These variance estimates are combined in a loss function to down-weight uncertain samples.

III. IV-PPO

Following the IV-RL framework proposed by Mai et al. [1], this work adapts uncertainty-aware RL to an on-policy setting using PPO [3]. The IV-PPO implementation extends a baseline critic ensemble by incorporating four key components: (A) a variance prediction head to model aleatoric uncertainty, (B) a target ensemble to estimate epistemic uncertainty, (C) Batch Inverse-Variance (BIV) weighting to reduce the influence of high-uncertainty samples, (D) Loss Attenuation (LA) to learn variance predictions robustly, and (E) a combined optimization objective that balances both weighting schemes. Each component and its role in modifying the PPO update mechanism are detailed in the following subsections.

A. Variance Head for Aleatoric Uncertainty

To model state-dependent environmental noise, each PPO critic network is augmented with a variance prediction head. The architecture consists of shared hidden layers (two layers, 256 units each, ReLU activation) followed by two output heads: one for the value estimate $V(s)$ and one for log-variance. The predicted aleatoric variance is obtained via:

$$\sigma_{\text{aleatoric}}^2(s) = \exp(\log_var(s)) \quad (1)$$

where the exponential ensures positivity and numerical stability. During policy rollouts and action selection, only the value head is queried; the variance head is activated only during critic training to enable uncertainty-aware updates.

B. Target Ensemble for Epistemic Uncertainty

Epistemic uncertainty is estimated via ensemble disagreement. $K = 5$ independently initialized critic networks (see Table I) are maintained, each with a corresponding target network updated via soft (Exponential Moving Average) updates with rate $\tau = 0.005$:

$$\theta_k^{\text{target}} \leftarrow \tau \theta_k + (1 - \tau) \theta_k^{\text{target}} \quad (2)$$

where θ_k represents the parameters of the shared layers and value head. The slow update rate ensures stable epistemic estimates during training.

Epistemic variance is computed from target ensemble disagreement on next-state values:

$$\sigma_{\text{epistemic}}^2(s') = \frac{1}{K} \sum_{k=1}^K (V_k^{\text{target}}(s') - \bar{V}^{\text{target}}(s'))^2 \quad (3)$$

where $\bar{V}^{\text{target}}(s') = \frac{1}{K} \sum_{k=1}^K V_k^{\text{target}}(s')$ is the mean prediction.

A key design asymmetry is used: Aleatoric uncertainty is predicted on the current state s (using online networks), while epistemic uncertainty is measured on the next state s' (using target networks). This prevents self-reinforcing feedback loops where high epistemic uncertainty in bootstrapped targets destabilizes aleatoric predictions.

C. Batch Inverse-Variance (BIV) Weighting

Total TD target uncertainty per sample i is defined as the sum of the two components:

$$\sigma_{\text{total},i}^2 = \sigma_{\text{aleatoric}}^2(s_i) + \sigma_{\text{epistemic}}^2(s'_i) \quad (4)$$

For each transition in a minibatch B , total uncertainty is then converted into a normalized inverse-variance weight:

$$w_i = \frac{1}{\sigma_{\text{total},i}^2 + \xi}, \quad \tilde{w}_i = \frac{w_i}{\sum_{j \in B} w_j} \quad (5)$$

where $\xi > 0$ is a small regularization constant that prevents extremely large weights when $\sigma_{\text{total},i}^2$ is near zero. Using these normalized weights, the BIV-weighted critic regression term for ensemble member k takes the form:

$$L_{\text{BIV}} = \sum_{i \in B} \tilde{w}_i (y_i - V_k(s_i))^2 \quad (6)$$

where $y_i = r_i + \gamma \bar{V}^{\text{target}}(s'_i)$ is the one-step TD target formed using the mean target-ensemble value. The BIV loss is essentially a weighted MSE loss where samples with higher total uncertainty receive lower weights, making the network focus on learning from more reliable supervision signals.

In implementation, $\tilde{w}_i$ is detached from the computation graph so that gradients do not propagate through the weighting computation. This prevents a degenerate solution in which the network reduces the weighted loss by artificially inflating predicted variances to shrink $\tilde{w}_i$, rather than improving value accuracy. Detaching enforces that uncertainty influences learning only by reweighting samples, not by providing a gradient shortcut.

BIV weighting can become overly concentrated on a small subset of transitions, effectively reducing the amount of information used per update. To monitor this, ξ can be adapted online to maintain a minimum effective batch size (e.g., 48 for batch size 64):

$$\text{BS}_{\text{eff}} = \frac{1}{\sum_{i \in B} \tilde{w}_i^2} \quad (7)$$

which equals the true batch size under uniform weights and decreases as weights become more skewed. Following Mai et al.[1], when BS_{eff} falls below the threshold, ξ is increased to reduce weight skewness and retain more effective samples. In this paper’s default configuration, ξ is fixed (`dynamic_xi = False`). This was chosen for simplicity and to prioritize fixed hyperparameters that generalize across all four test environments (Pendulum, LunarLander, HalfCheetah, Walker2d) without environment-specific tuning. The dynamic adaptation adds complexity without demonstrated benefit when a single ξ value works robustly across diverse task types.

D. Loss Attenuation (LA) for Learning Variance

While BIV weighting uses uncertainty to down-weight samples, the variance head must be trained in a way that discourages degenerate solutions. In particular, if the variance

head were trained only through its downstream effect on BIV weighting, it could learn to arbitrarily manipulate weights rather than represent true stochasticity. IV-PPO therefore learns aleatoric uncertainty by casting value regression as heteroscedastic Gaussian likelihood:

$$y_i \sim \mathcal{N}(V(s_i), \sigma_{\text{aleatoric}}^2(s_i)) \quad (8)$$

Dropping constants and substituting log-variance, the resulting LA (negative log-likelihood) objective is:

$$\begin{aligned} L_{\text{LA}} &= \frac{1}{2} \log \sigma_{\text{aleatoric}}^2(s_i) + \frac{1}{2\sigma_{\text{aleatoric}}^2(s_i)} (y_i - V(s_i))^2 \\ &= \frac{1}{2} \log_var_i + \frac{1}{2} \exp(-\log_var_i) (y_i - V(s_i))^2 \end{aligned} \quad (9)$$

typically averaged over the minibatch. This loss has two coupled effects: (1) the $\frac{1}{2} \log \sigma^2$ term regularizes against uniformly large variances, while (2) the precision-weighted error term $\frac{(y-V)^2}{2\sigma^2}$ attenuates the penalty for large residuals when the environment is inherently noisy. Thus, the LA objective is minimized when the predicted variance tracks the squared residual, encouraging the variance head to assign high variance only to persistently noisy (hard-to-predict) regions and low variance to predictable regions, consistent with heteroscedastic RL targets.

This behaviour is made explicit by the gradient with respect to the `log_var`:

$$\frac{\partial L_{\text{LA}}}{\partial \log_var} = \frac{1}{2} - \frac{1}{2} \exp(-\log_var) (y - V)^2 \quad (10)$$

so large squared errors increase `log_var` (raising predicted variance), while small errors decrease `log_var` (tightening the variance). This provides a direct mechanism for aligning predicted aleatoric uncertainty with observed outcome noise.

E. Combined IV-RL Loss for Critic Update

IV-PPO trains the critic with a single objective that blends uncertainty-based sample reweighting and heteroscedastic likelihood learning by taking a convex combination of the BIV-weighted MSE L_{BIV} and LA term L_{LA} :

$$L_{\text{IV-RL}} = \lambda \cdot L_{\text{BIV}} + (1 - \lambda) \cdot L_{\text{LA}} \quad (11)$$

This introduces a trade-off between emphasizing reliable samples (via BIV) and learning calibrated aleatoric variance (via LA); unless otherwise stated, experiments use $\lambda = 0.5$ to weight both terms equally, though λ can be tuned per environment.

IV. RESULTS

IV-PPO was evaluated against two baselines across four continuous control environments (see Table II): PPO, the standard single-critic variant, and EnsemblePPO, which uses a $K = 5$ critic ensemble, while IV-PPO combines the ensemble critic with a variance head to model both uncertainty types.

All experiments use 5 random seeds per method (see Fig. 1). Final performance is computed as the mean return over the last

TABLE I
HYPERPARAMETER CONFIGURATION

<i>Parameter</i>	<i>Default</i>	<i>Description</i>
num_nets	5	Number of ensemble members (K)
tau	0.005	Soft update rate for target networks (very slow update)
xi	1.0	BIV weight regularization
dynamic_xi	False	Whether to adapt xi to maintain effective batch size
minimal_eff_bs	48	Target effective batch size (if dynamic_xi=True)
lambda_div	0.5	Weight for BIV loss vs LA loss

TABLE II
ENVIRONMENTS OVERVIEW

<i>Environment</i>	<i>State</i>	<i>Action</i>	<i>Characteristics</i>
Pendulum-v1 [16]	3	1	Deterministic dynamics, homoscedastic rewards
LunarLanderCont.-v2 [17]	8	2	Stochastic wind, heteroscedastic transitions
HalfCheetah-v4 [18]	17	6	High-dimensional, contact-rich dynamics
Walker2d-v4 [19]	17	6	Balance task with discrete contact events

100 episodes (see Table III, where bold indicates best gain over PPO).

A. Pendulum-v1 (Deterministic, homoscedastic)

Pendulum-v1 is effectively deterministic with a uniform reward structure [16], so it serves primarily as a control case for checking whether uncertainty-aware training introduces regressions rather than as a strong test of learning under stochasticity. In Fig. 2, across most of training, EnsemblePPO (orange) maintains the highest (least negative) mean return, while PPO (blue) tends to be lowest; IV-PPO (green) sits in between and is often closer to EnsemblePPO than PPO. All three methods show large, persistent variability (wide shaded bands) and frequent oscillations, so the curves largely overlap despite the mean ordering.

In Table III, PPO and IV-PPO achieve nearly-identical outcomes (PPO: -1396.5 ± 195.0 ; IV-PPO: -1398.1 ± 160.0 , -0.1%), whereas EnsemblePPO performs better (-1272.4 ± 175.7 , $+8.9\%$ over PPO). Interpreting this result, the lack of IV-PPO gains is expected when there is minimal heteroscedasticity to exploit; the environment therefore functions as a “no-regret” sanity check rather than evidence for benefits under stochasticity.

B. LunarLanderContinuous-v2 (Stochastic wind, heteroscedastic transitions)

LunarLanderContinuous-v2 introduces stochastic wind and contact-rich dynamics [17], producing state-dependent vari-

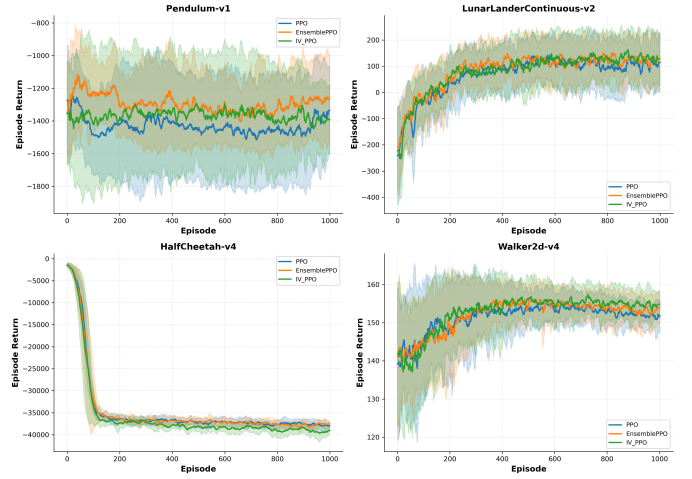


Fig. 1. Learning Curves Across Environments. Mean episode return over training for PPO, Ensemble PPO, and IV-PPO across four benchmark tasks. Shaded regions represent 95% confidence intervals using t-distribution across random seeds.

TABLE III
PERFORMANCE COMPARISON ACROSS ENVIRONMENTS

<i>Env.</i>	<i>PPO</i>	<i>IV-PPO</i>	<i>Ensemble PPO</i>
Pendulum-v1	-1396.5 ± 195.0	-1398.1 ± 160.0 (-0.1%)	-1272.4 ± 175.7 ($+8.9\%$)
LunarLanderCont.-v2	106.1 ± 24.8	129.3 ± 29.9 ($+21.9\%$)	121.8 ± 21.2 ($+14.8\%$)
HalfCheetah-v4	-37744.6 ± 434.3	-39197.1 ± 580.1 (-3.8%)	-37720.5 ± 277.4 ($+0.1\%$)
Walker2d-v4	151.9 ± 0.6	154.7 ± 1.4 ($+1.8\%$)	153.0 ± 1.7 ($+0.7\%$)

ability in outcomes that better matches the IV-RL motivation. In Fig. 3, all methods improve quickly in first ~ 200 episodes, then settle into a noisy plateau. EnsemblePPO and IV-PPO are generally slightly above PPO after convergence, but the shaded regions remain wide and heavily overlapping, indicating substantial run-to-run variance even late in training.

In Table III, IV-PPO demonstrates the strongest final performance at 129.3 ± 29.9 versus PPO’s 106.1 ± 24.8 ($+21.9\%$), also exceeding EnsemblePPO at 121.8 ± 21.2 . A plausible interpretation is that inverse-variance weighting reduces the influence of high-noise TD targets induced by wind, improving learning efficiency in a heteroscedastic setting; however, the standard deviations (std) remain substantial, so stronger stability claims would require more seeds and/or longer runs.

C. HalfCheetah-v4 (High-dimensional, contact-rich)

HalfCheetah-v4 is high-dimensional and contact-rich [18], and the learning curves exhibit a steep early drop in ~ 100 – 150 episodes followed by a plateau with relatively small separation between methods. Fig. 4 shows that IV-PPO (green) trends slightly lower (more negative) than PPO and EnsemblePPO across much of the plateau, and the confidence bands overlap but the mean gap persists visually.

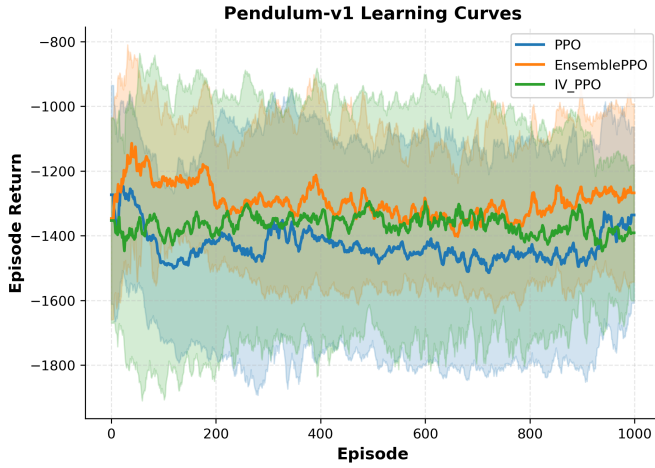


Fig. 2. Pendulum-v1 Learning Curves.

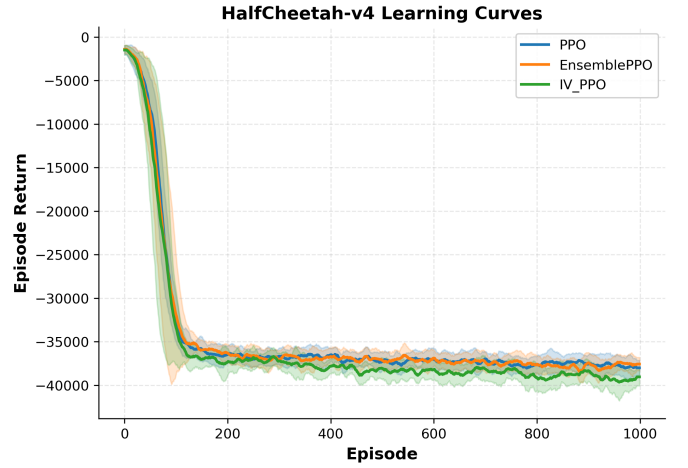


Fig. 4. HalfCheetah-v4 Learning Curves.

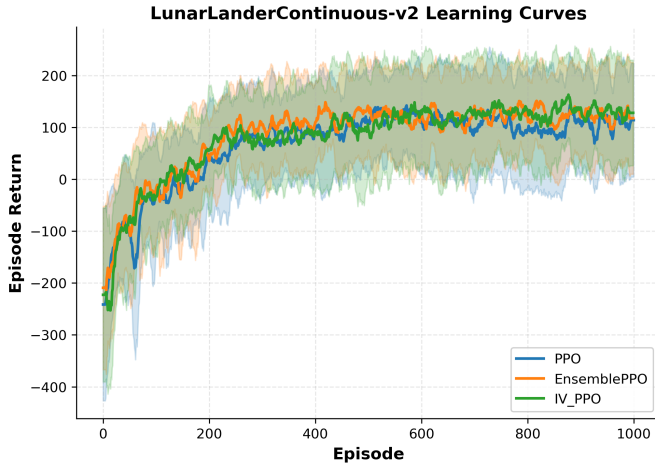


Fig. 3. LunarLanderContinuous-v2 Learning Curves.

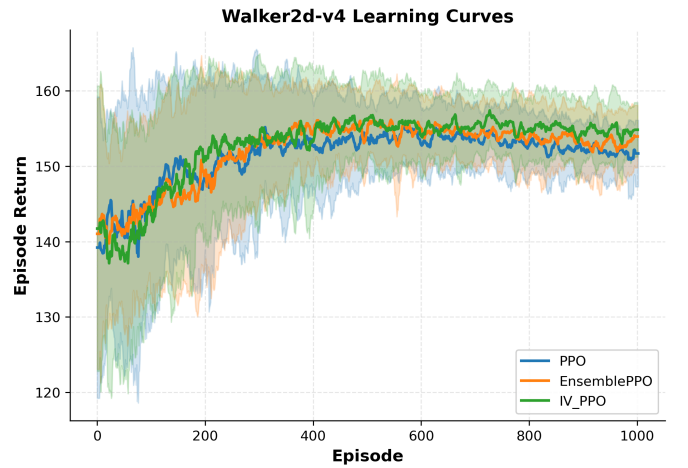


Fig. 5. Walker2d-v4 Learning Curves.

Table III reveals that PPO and EnsemblePPO are essentially tied (PPO: -37744.6 ± 434.3 ; EnsemblePPO: -37720.5 ± 277.4 , $+0.1\%$), while IV-PPO underperforms (-39197.1 ± 580.1 , -3.8%). A reasonable interpretation is that, under this configuration, inverse-variance weighting may become overly conservative or miscalibrated in a difficult, contact-driven regime, reducing the impact of informative but noisy transitions. The main limitation here is sensitivity to training budget and hyperparameters in MuJoCo-style locomotion.

D. Walker2d-v4 (Locomotion with contact events)

Walker2d-v4 is a locomotion and balance task with contact events [19], but it appears less heteroscedastic than LunarLander under the current training setup. The learning curves rise and then compress into a narrow band, suggesting relatively stable convergence across seeds. In Fig 5, returns increase steadily during the early phase and then stabilize around the mid-150s. IV-PPO (green) is typically the highest curve after ~ 200 episodes, with EnsemblePPO (orange) close behind and PPO (blue) slightly lower toward the end; however, the

uncertainty bands overlap, so differences are modest relative to variability.

Table III suggests a modest improvement for IV-PPO (151.9 ± 0.6 , $+1.8\%$), with EnsemblePPO in between (153.0 ± 1.7 , $+0.7\%$). Given the small effect size and overlapping uncertainty regions, this result is best treated as limited evidence that uncertainty-weighted updates help slightly in this domain, rather than a definitive improvement in either stability or sample efficiency.

V. DISCUSSION

A. Sample Efficiency and Learning Stability

Across environments, the results suggest that IV-PPO's benefits are not uniform; they depend on whether the task exposes meaningful, state-dependent stochasticity that the inverse-variance mechanism can exploit. Under the sample efficiency criterion, the learning curves indicate that all three methods typically make their largest gains early (first few hundred episodes), after which they plateau, but

LunarLanderContinuous-v2 is the clearest case where IV-PPO translates uncertainty modeling into improved returns. Under the learning stability criterion, however, the cross-seed variability is mixed: IV-PPO does not consistently reduce variance (e.g., in LunarLander and Walker2d, IV-PPO’s std is larger than PPO’s indicated in Table III), which implies that in this on-policy PPO adaptation, uncertainty-weighting can improve final performance without guaranteeing more consistent runs.

B. Limitations

- 1) *Stability evidence is inconclusive at 5 seeds:* With only five random seeds, the shaded confidence bands remain broad in some environments, and the std comparisons show that IV-PPO sometimes increases cross-seed variance (e.g., LunarLander, Walker2d, HalfCheetah) rather than reliably decreasing it.
- 2) *Task stochasticity mismatch:* The strongest improvement appears where stochastic effects are salient (LunarLanderContinuous-v2), while contact-rich MuJoCo locomotion (HalfCheetah/Walker2d) may not present uncertainty in the form that inverse-variance weighting is designed to exploit, limiting transferability of the core claim.
- 3) *No direct uncertainty visualization:* While the method estimates epistemic and aleatoric variance, these quantities were not logged during training, preventing direct analysis of how uncertainty evolves or correlates with learning progress.
- 4) *Hyperparameter sensitivity:* IV-PPO introduces additional hyperparameters (ξ, λ_{BIV}, τ) that may require environment-specific tuning for optimal results.
- 5) *Computational overhead:* IV-PPO requires $\sim 1.2-1.4\times$ the training time of PPO due to ensemble forward passes and variance computation.

VI. CONCLUSION

Overall, the results support a qualified conclusion: uncertainty-aware critic training via IV-PPO can improve learning performance - and in some cases learning speed - when stochasticity meaningfully corrupts TD supervision, but the benefits do not transfer uniformly across continuous-control domains. The strongest positive evidence appears in LunarLanderContinuous-v2, where IV-PPO achieves higher final returns than both PPO and EnsemblePPO and maintains a higher mean trajectory over training, consistent with the intended effect of down-weighting high-uncertainty updates. In contrast, in environments where stochasticity is weak (Pendulum) or where optimization/contact dynamics dominate (HalfCheetah), the inverse-variance mechanism provides limited benefit and may be harmful under fixed hyperparameters. Given the observed cross-seed variability and the limited number of seeds, broader claims about improved stability require further evaluation (more seeds, longer budgets). Future work should focus on (1) environment-specific tuning of ξ, λ , and ensemble design, (2) direct uncertainty calibration analyses (e.g., predicted variance vs. realized TD error), and (3) investigating why contact-rich locomotion tasks may require different uncertainty handling than wind-driven or explicitly stochastic dynamics.

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